

Class 3

Ground state problem

$$H|\psi_0\rangle = E_0|\psi_0\rangle$$

$$|\psi_0\rangle \in \mathcal{H} = (\mathcal{H}_1)^{\otimes N} = \mathcal{H}_1 \otimes \mathcal{H}_1 \otimes \dots \otimes \mathcal{H}_1$$

$$\dim(\mathcal{H}_1) = d$$

$$\dim(\mathcal{H}) = d^N$$

- examples:
- spin systems
 - fermion models

We saw that E.D. breaks down, if

$$d^N \gtrsim 10^6 - 10^8$$



↑ accessible if symmetries are used.

$$N \lesssim \frac{\log(10^8)}{\log(d)} \sim 27 \quad (d=2)$$

$$N \lesssim \frac{\log(10^8)}{\log(d)} \sim 17 \quad (d=3)$$

APPROXIMATION is necessary

Idea: coarse graining & RG

- do not keep all microscopic states, only the "important" ones
- Chop the system into smaller parts, and keep selected states of the parts
- Iterate, and hope in convergence

Simplest approach: Block-RG

Jullien et al. PRB 18, 3568 (1978)

Example system: transverse field Ising model in 1D

$$H = -J \sum_i \sigma_i^x \sigma_{i+1}^x + h \sum_i \sigma_i^z$$

Form 2-spin blocks

$$i \rightarrow j, p$$

$$j = \text{int}(\frac{i}{2})$$

$$p = \text{mod}(i, 2) \in \{0, 1\}$$

$$H = \sum_j (H_j + V_{j,j+1})$$

$$H_j = -J \sigma_{2j}^x \sigma_{2j+1}^x + h (\sigma_{2j}^z + \sigma_{2j+1}^z)$$

$$V_{j,j+1} = -J \sigma_{2j+1}^x \sigma_{2j+2}^x$$

1) Diagonalize $H_j = \begin{pmatrix} 2h & -J \\ & -J & -J \\ -J & & -2h \end{pmatrix}$ $\begin{matrix} |\uparrow\uparrow\rangle \\ |\uparrow\downarrow\rangle \\ |\downarrow\uparrow\rangle \\ |\downarrow\downarrow\rangle \end{matrix}$

α	E_α
0	$-\sqrt{J^2+4h^2}$
1	$-J$
2	$+J$
3	$+\sqrt{J^2+4h^2}$

2) Keep only the two lowest states of H_j

$$E_0 = -\sqrt{J^2+4h^2} \quad |0\rangle = \frac{1}{\sqrt{A^2+B^2}} (A|\uparrow\uparrow\rangle + B|\downarrow\downarrow\rangle)$$

$$A = J$$

$$B = 2h + \sqrt{J^2+4h^2}$$

$$E_1 = -J \quad |1\rangle = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)$$

3) Restrict 2-spin blocks to $|0\rangle, |1\rangle$

$$|\tilde{\uparrow}\rangle \equiv |1\rangle \quad |\tilde{\downarrow}\rangle \equiv |0\rangle$$

$$H_j = E_0 |\tilde{\downarrow}\rangle\langle\tilde{\downarrow}| + E_1 |\tilde{\uparrow}\rangle\langle\tilde{\uparrow}| = \frac{E_0+E_1}{2} \underbrace{(|\tilde{\uparrow}\rangle\langle\tilde{\uparrow}| + |\tilde{\downarrow}\rangle\langle\tilde{\downarrow}|)}_{\tilde{1}_j} + \frac{E_1-E_0}{2} \underbrace{(|\tilde{\uparrow}\rangle\langle\tilde{\uparrow}| - |\tilde{\downarrow}\rangle\langle\tilde{\downarrow}|)}_{\tilde{\sigma}_j^z}$$

$$\langle\tilde{\uparrow}|\sigma_{2j}^x|\tilde{\uparrow}\rangle = 0 = \langle\tilde{\downarrow}|\sigma_{2j}^x|\tilde{\downarrow}\rangle$$

$$\langle\tilde{\downarrow}|\sigma_{2j}^x|\tilde{\uparrow}\rangle = \frac{1}{\sqrt{2}\sqrt{A^2+B^2}} (A\langle\uparrow\uparrow| + B\langle\downarrow\downarrow|) \underbrace{\sigma_{2j}^x (|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)}_{|\downarrow\downarrow\rangle + |\uparrow\uparrow\rangle} = \frac{A+B}{\sqrt{2}\sqrt{A^2+B^2}}$$

||

$$\langle\tilde{\uparrow}|\sigma_{2j}^x|\tilde{\downarrow}\rangle$$

Interestingly, the matrix elements of σ_{2j+1}^x are exactly the same

$$\rightarrow \sigma_{2j}^x \sigma_{2j+1}^x \rightarrow \frac{A+B}{\sqrt{2}\sqrt{A^2+B^2}} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \frac{A+B}{\sqrt{2}\sqrt{A^2+B^2}} \tilde{\sigma}_j^x$$

$$\Rightarrow H = \sum_j \underbrace{\frac{E_0+E_1}{2} \tilde{1}_j + \frac{E_1-E_0}{2} \tilde{\sigma}_j^z}_{H_j} - J \underbrace{\frac{(A+B)^2}{2(A^2+B^2)} \tilde{\sigma}_j^x \tilde{\sigma}_{j+1}^x}_{V_{j,j+1}}$$

This has the same form as before, but

- $C_j = \frac{\varepsilon_0 + \varepsilon_1}{2} = \frac{1}{2} (-\sqrt{g^2 + 4h^2} - g)$ constant

- $\tilde{h} = \frac{\varepsilon_1 - \varepsilon_0}{2} = \frac{1}{2} (\sqrt{g^2 + 4h^2} - g) = \frac{1}{2} h \left(\sqrt{\frac{g^2}{h^2} + 4} - \frac{g}{h} \right)$

- $\tilde{g} = \frac{1}{2} g \frac{(g + 2h + \sqrt{g^2 + 4h^2})^2}{g^2 + (2h + \sqrt{g^2 + 4h^2})^2} = \frac{1}{2} g \cdot \frac{(\frac{g}{h} + 2 + \sqrt{\frac{g^2}{h^2} + 4})^2}{\frac{g^2}{h^2} + (2 + \sqrt{\frac{g^2}{h^2} + 4})^2}$

$\Rightarrow$ RG transformation

$$\frac{\tilde{g}}{\tilde{h}} = \frac{g}{h} \cdot \frac{(\frac{g}{h} + 2 + \sqrt{\frac{g^2}{h^2} + 4})^2}{(\sqrt{\frac{g^2}{h^2} + 4} - \frac{g}{h})(\frac{g^2}{h^2} + (2 + \sqrt{\frac{g^2}{h^2} + 4})^2)}$$

$$\tilde{x} = x \cdot \frac{(x + 2 + \sqrt{x^2 + 4})^2}{(\sqrt{x^2 + 4} - x)(x^2 + (2 + \sqrt{x^2 + 4})^2)}$$

$\Delta(x)$

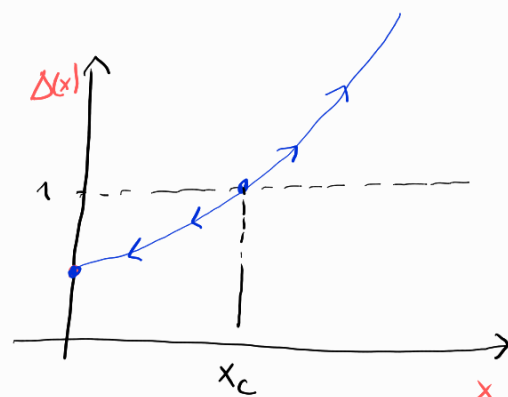
Proposed critical coupling:

$$\Delta(x_c) = 1$$

$$x_c = \left(\frac{g}{h}\right)_c \approx 0.78$$

$$\Rightarrow \boxed{h_c = 1.28 g}$$

WRONG !!



$$h_c = g$$

(exact solution by Jordan-Wigner transformation)

How to do it better?

BRG algorithm 2

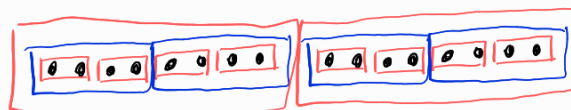
• Form π -blocks

$$j = \text{int}(i/\pi)$$

$$H = \sum_j (H_j + V_{j,j+1} + \text{const})$$

$$H_j = \sum_{p=0}^{\pi-1} h \cdot \sigma_{\tau j+p}^z + \sum_{p=0}^{j-2} -g \sigma_{\tau j+p}^x \sigma_{\tau j+p+1}^x$$

• We diagonalized always $\pi=2$ spins, and then truncated



$\hookrightarrow$ we can avoid many truncations, by using $\pi=4, 8, 16$

$\rightarrow$ we can use other π values too.

$$V_{j,j+1} = -g \sigma_{\tau j+\pi-1}^x \sigma_{\tau(j+1)}^x$$

- Diagonalize H_j & keep the 2 lowest states

- $\tilde{H}_j = C \cdot \mathbb{I} + \tilde{h} \tilde{\sigma}_j^z$ $\tilde{h} = \frac{\epsilon_1 - \epsilon_0}{2}$

- Calculate matrix elements of $\sigma_{r_j}^x$ (first spin of block)

$\sigma_{r_j+r-1}^x$ (last spin of block)

(these will be equal, $\sum_j \tilde{\sigma}_j^x$)

- $\tilde{J} = J \cdot g$

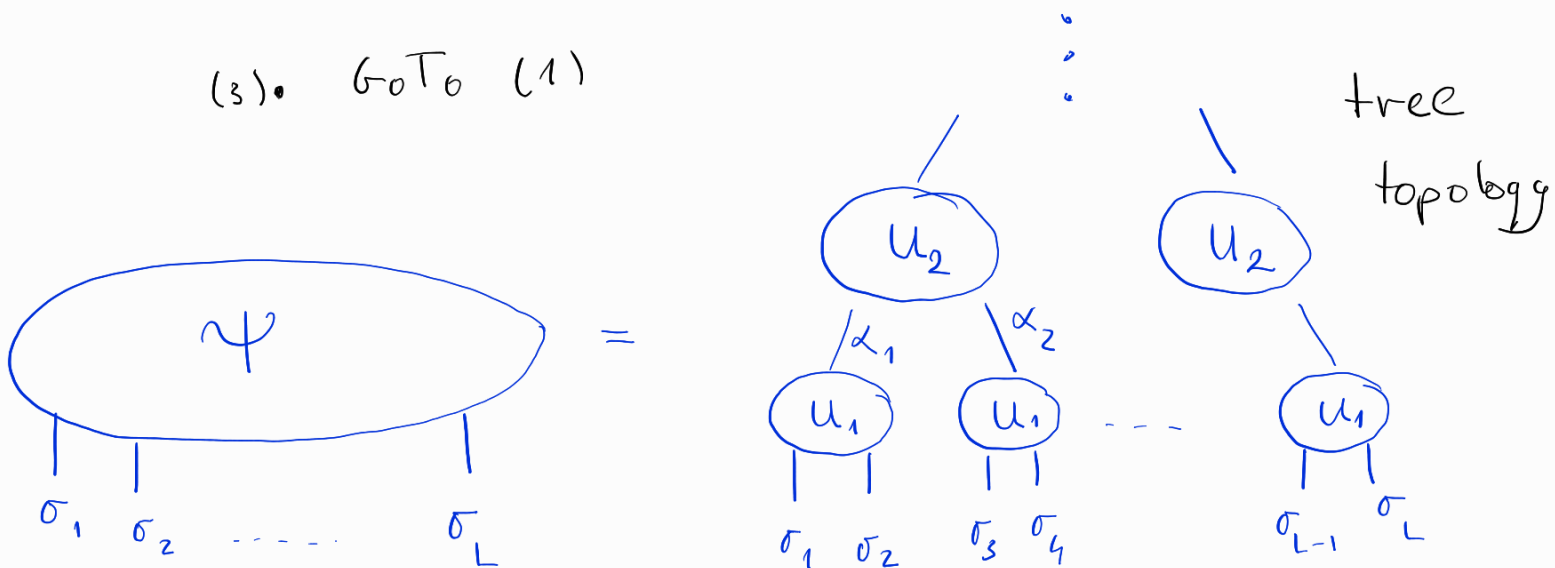
- Reformulate the result as a Tensor Network State

$\tau=2$

- (1) • Diagonalize H_j , and form $|\alpha\rangle = \sum_{\sigma_1, \sigma_2} U_{\sigma_1 \sigma_2}^\alpha |\sigma_1\rangle |\sigma_2\rangle$

- (2) • Rewrite the Hamiltonian in the truncated basis

- (3) • Go To (1)



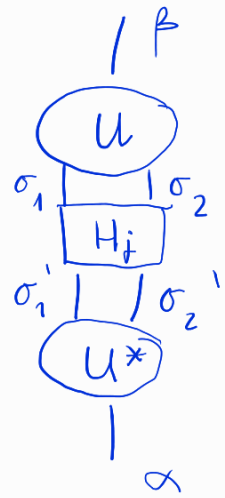
This is a Tensor Network!

- Graph rules: summation for contracted legs

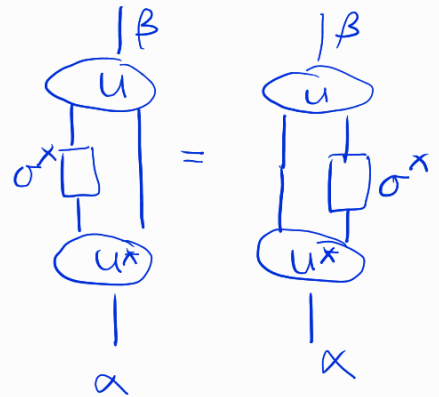
- $\dim(\alpha) = 2$

BRG transformation

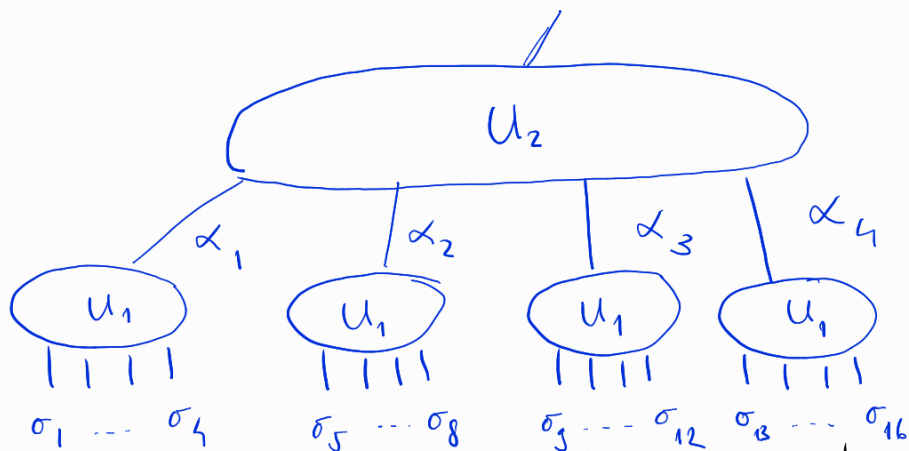
$$\left[C \cdot 11 + \tilde{h} \tilde{\sigma}^z \right]_{\alpha\beta} = \langle \alpha | H_j | \beta \rangle$$



$$\left[\tilde{\sigma}^x \right]_{\alpha\beta} = \langle \alpha | \sigma_2^x | \beta \rangle = \langle \alpha | \sigma_1^x | \beta \rangle$$



$$\tau > 2$$



• keeping $\dim(\alpha) = 2$ is a huge restriction

• We may increase it: $\dim(\alpha) = M$

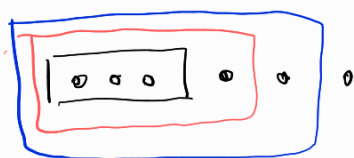
$$\Rightarrow \dim(\mathcal{H}_j) = M^\tau$$

↳ we lose the Ising model form of the Hamiltonian

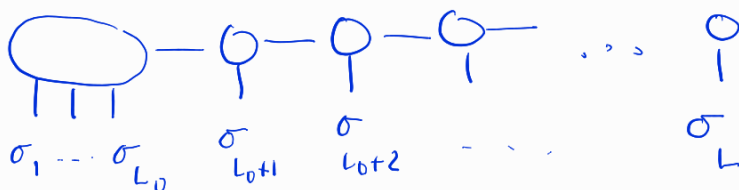
↳ RG: $H_0 \rightarrow \tilde{H}_1 \rightarrow \tilde{H}_2 \rightarrow \dots \rightarrow \tilde{H}_\infty$

NRG

- Diagonalize a 1D chain iteratively
- First: short chain $L_0 \ll L$
 - L_0 solve with ED
 - L_0 Keep only $M \leq 2^{L_0}$ lowest states
- * • Add 1-site to the chain
- Diagonalize and keep only M lowest states
- if $L_i < L$ goto *



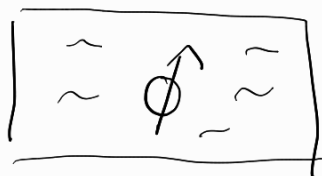
The result is a TNS again!



→ Poor performance for spin chains

- The new spin is always at the boundary
- We cannot correct initial errors later

→ Good performance in impurity models



Wilson chain (after mapping)

$$H = \epsilon_d n_d + U n_{\uparrow}^d n_{\downarrow}^d +$$

$$V \sum_{\sigma} \left(f_{0\sigma}^{\dagger} d_{\sigma} + d_{\sigma}^{\dagger} f_{0\sigma} \right) +$$

$$\sum_{n=0, \sigma}^{\infty} \epsilon_n f_{n\sigma}^{\dagger} f_{n+1\sigma} + \lambda_n (f_{n\sigma}^{\dagger} f_{n+1\sigma} + \text{h.c.})$$

λ_n hopping decreases exponentially along the chain

Basic Idea of TNS

- Chop the system into subsystems
- Determine important states of subsystems
- Keep only those

What are the important states?

- Low-energy eigenstates of a part of H ?

↳ maybe OK, but not in general

- Let suppose, we know $|\psi_0\rangle$
two subsystems A & B

$$\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$$

$$|\psi_0\rangle = \sum_{ij} C_{ij} |i\rangle_A |j\rangle_B$$

Important states in A are those that are needed in expectation value calculations.

Reduced Density Matrix

$$\rho^A = \text{tr}_B |\psi_0\rangle\langle\psi_0| = \sum_{i,i'} \underbrace{\left(\sum_j C_{ij} C_{i'j}^* \right)}_{\rho^A_{ii'}} |i\rangle_A \langle i'|_A$$

$$\langle\psi_0| O^A |\psi_0\rangle = \text{tr}(\rho^A O^A)$$

Idea: diagonalize $\rho^A \Rightarrow p_1^A > p_2^A > \dots$ eigen values

↳ keep eigenstates with the M largest eigen values

$\Rightarrow$ Density Matrix RG